

Object-Oriented Programming Using Java

Introductions to Java

Contents

- About the Java Technology
- A Simple Java Program
- Creating, Compiling, and Executing a Java Program

References:

<https://docs.oracle.com/javase/tutorial/getStarted/index.html>

History

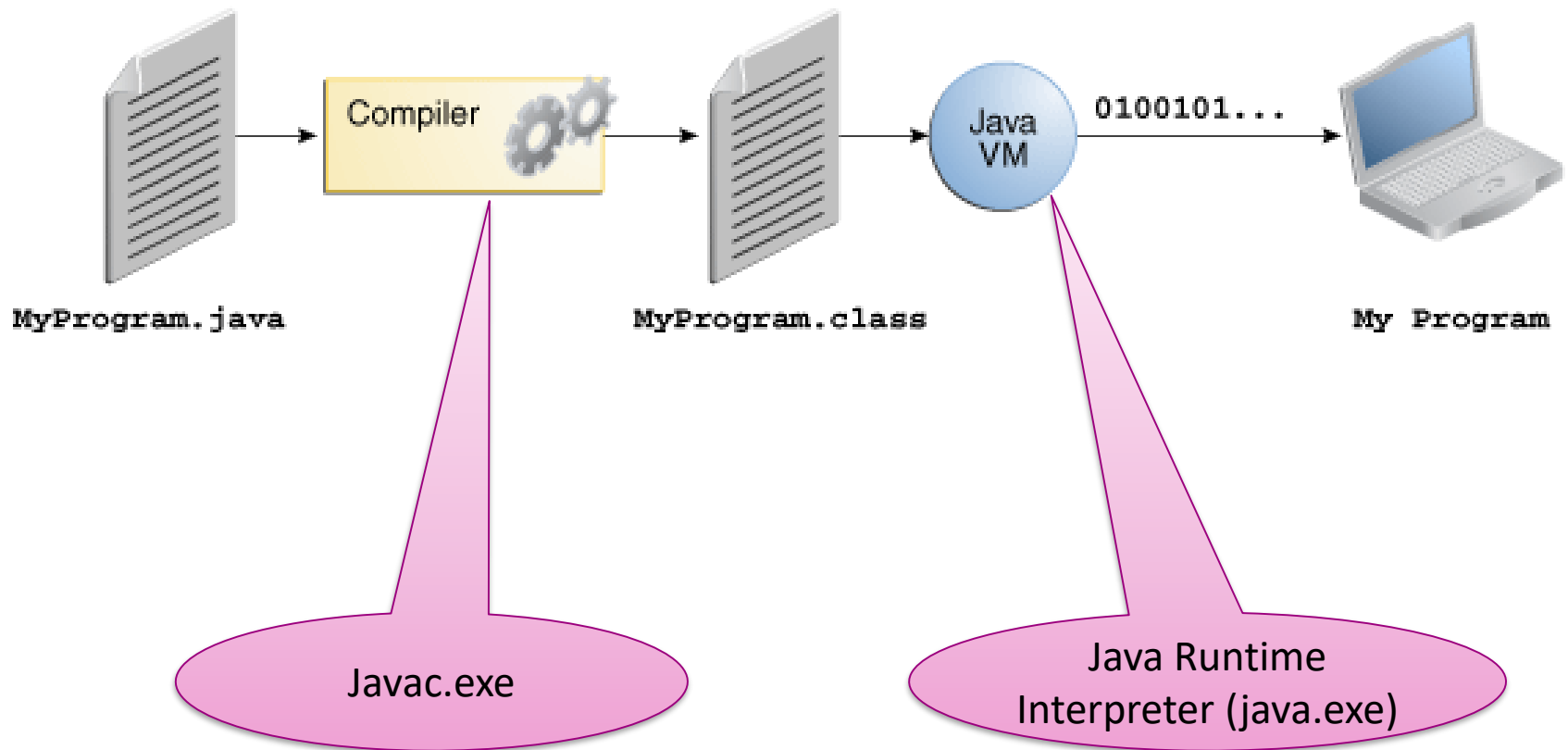
- 1990, James Gosling, Bill Joy, Patrick Naughton (Sun Microsystems) developed the **Oak language**
- 1993, Internet and Web were introduced, Sun change the Oak to an internet-development environment with a new project, named **Java**.



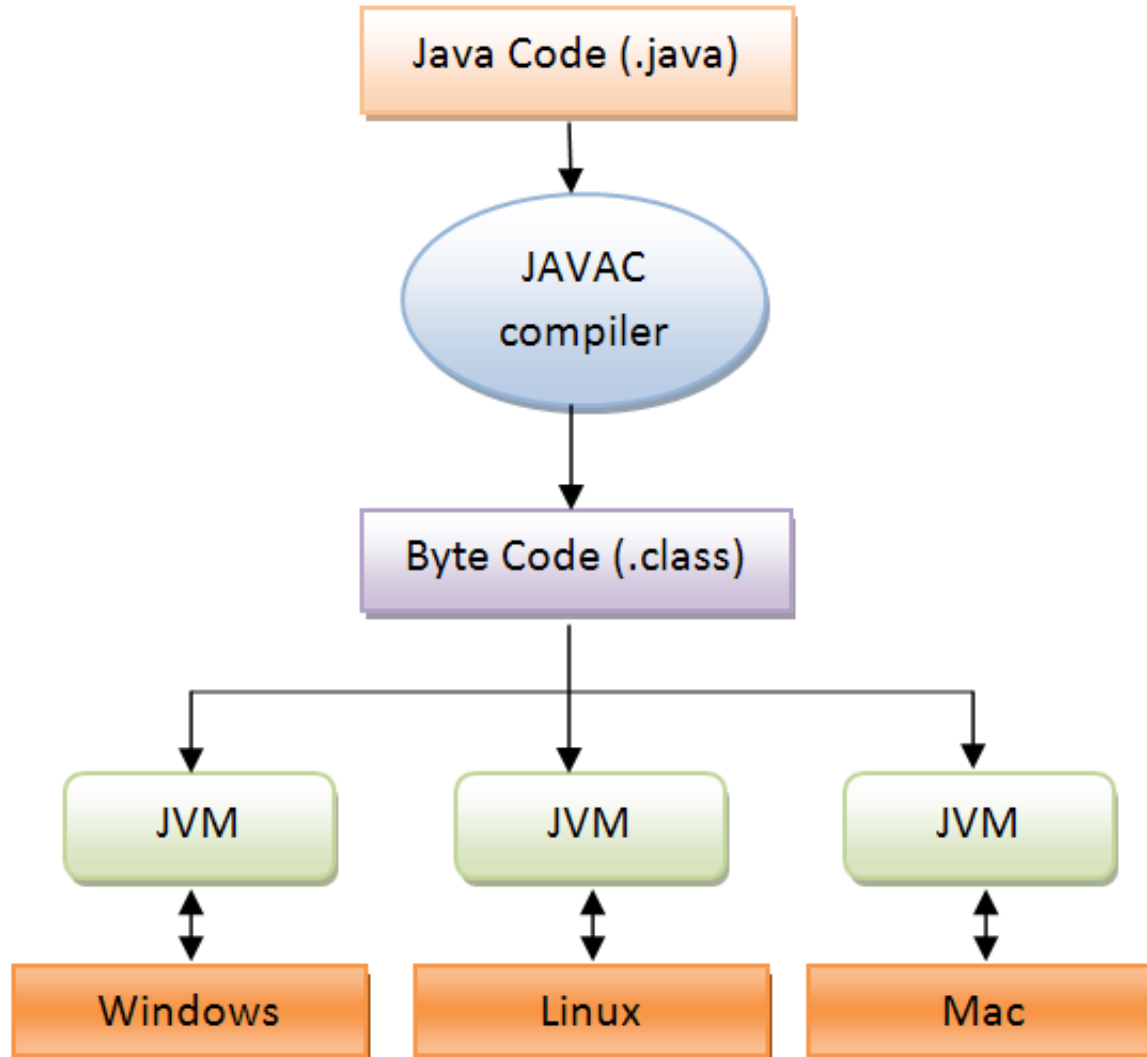
The Java Programming Language

- The Java programming language is a high-level language that can be characterized by:
 - Simple
 - **Object oriented**
 - Distributed
 - Multithreaded
 - Dynamic linking
 - Architecture neutral
 - Portable
 - High performance
 - Robust
 - Secure

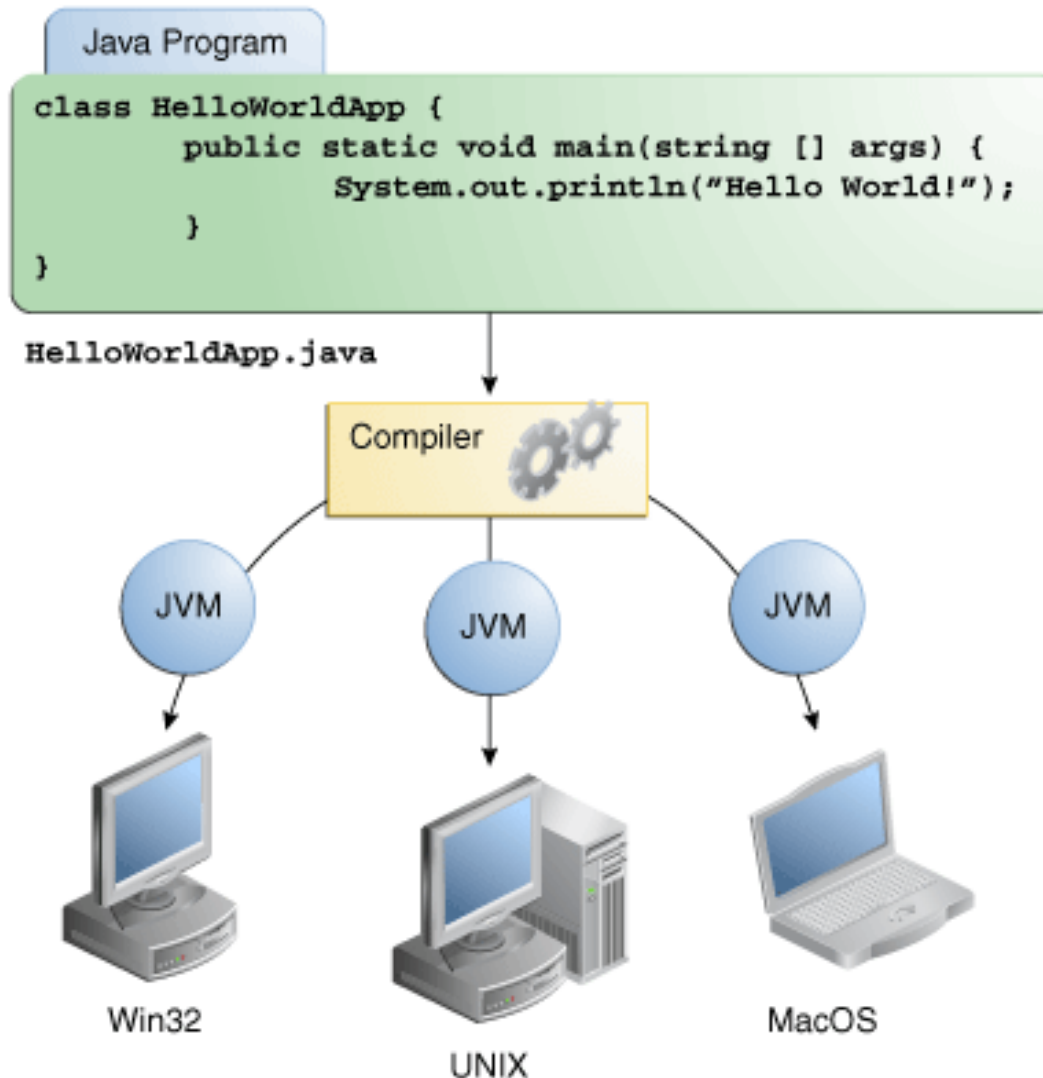
An overview of the software development process.



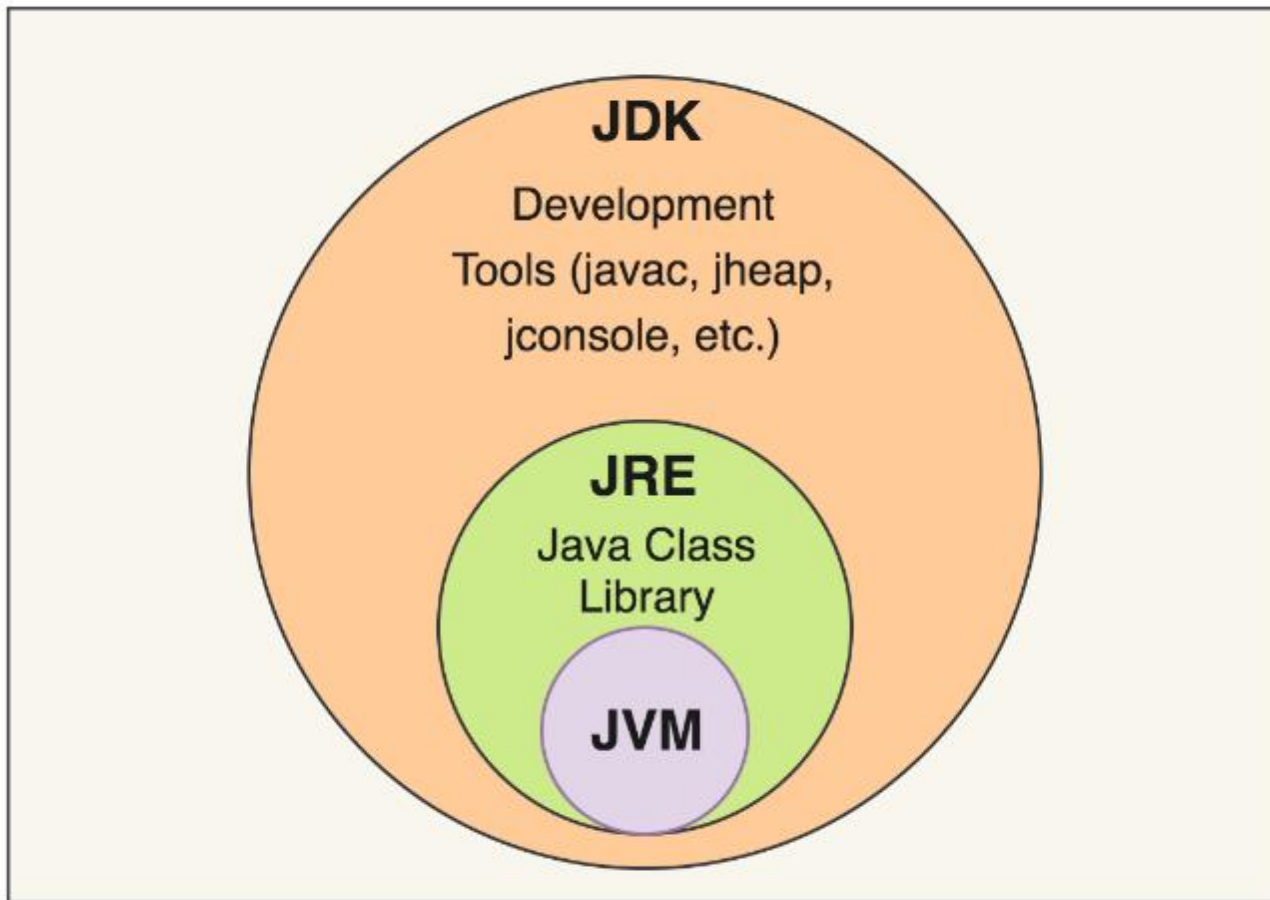
How can Java support platform-independence?



Write once, run anywhere



Differences between JDK, JRE and JVM



"Hello World!" for the NetBeans IDE

- The Java SE Development Kit (JDK)
<https://www.oracle.com/technetwork/java/javase/downloads/index.html>
- The NetBeans IDE (jdk8)
<https://netbeans.apache.org/download/index.html>
<https://netbeans.org/downloads/8.2/rc/>
- Apache NetBeans
<https://netbeans.apache.org/download/>

NetBeans IDE 8.2 Download

<https://netbeans.org/downloads/8.2/rc/>

NetBeans IDE Download Bundles

Supported technologies *	Java SE	Java EE	HTML5/JavaScript	PHP	C/C++	All
NetBeans Platform SDK	•	•				•
Java SE	•	•				•
Java FX	•	•				•
Java EE		•				•
Java ME						•
HTML5/JavaScript		•	•	•		•
PHP			•	•		•
C/C++					•	•
Groovy						•
Java Card™ 3 Connected						•
Bundled servers						
GlassFish Server Open Source Edition 4.1.1		•				•
Apache Tomcat 8.0.27		•				•

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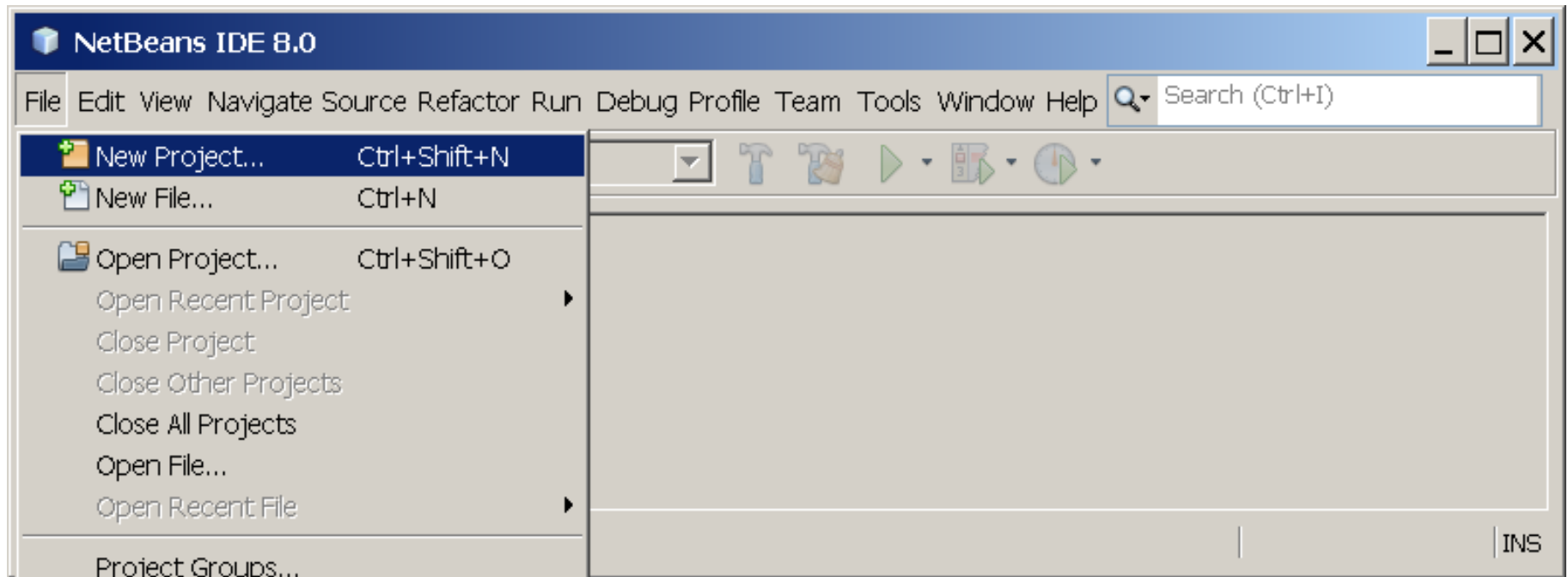
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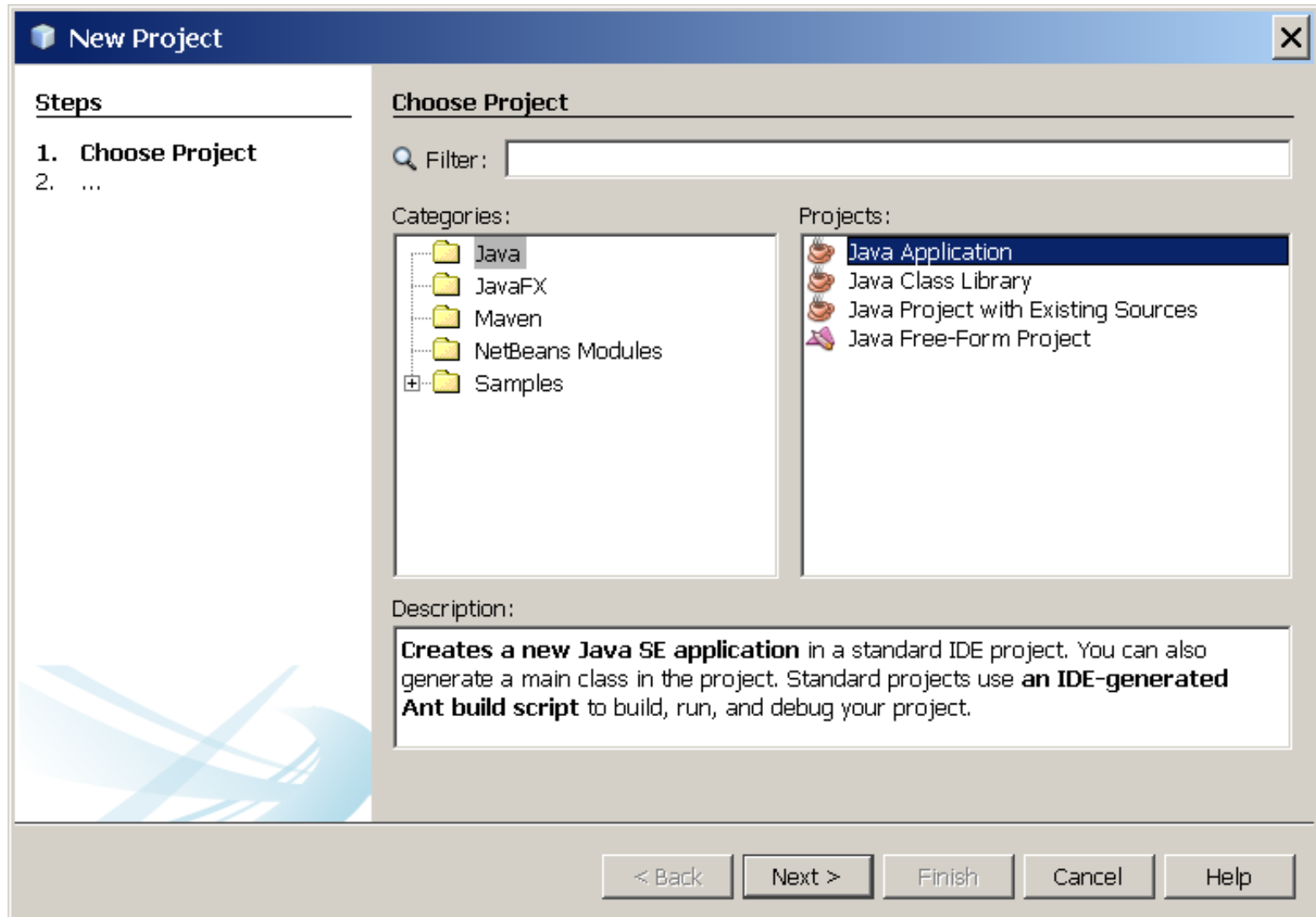
Creating Your First Application

1. Create an IDE project
2. Add code to the generated source file
3. Compile the source file into a *.class* file
4. Run the program

New Project



Choose Project page



Name and Location page

The screenshot shows the 'New Java Application' dialog box with the 'Name and Location' page selected. The 'Steps' list on the left shows '1. Choose Project' and '2. Name and Location'. The main area contains fields for 'Project Name' (Hello World App), 'Project Location' (C:\NetBeansProjects), and 'Project Folder' (C:\NetBeansProjects\Hello World App). There are 'Browse...' buttons for both the location and folder fields. A checkbox 'Use Dedicated Folder for Storing Libraries' is unchecked, with a 'Libraries Folder' field and 'Browse...' button below it. A note states: 'Different users and projects can share the same compilation libraries (see Help for details)'. A checkbox 'Create Main Class' is checked, with a text field containing 'helloworldapp.HelloWorldApp'. At the bottom are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

New Java Application

Steps

1. Choose Project
2. **Name and Location**

Name and Location

Project Name:

Project Location:

Project Folder:

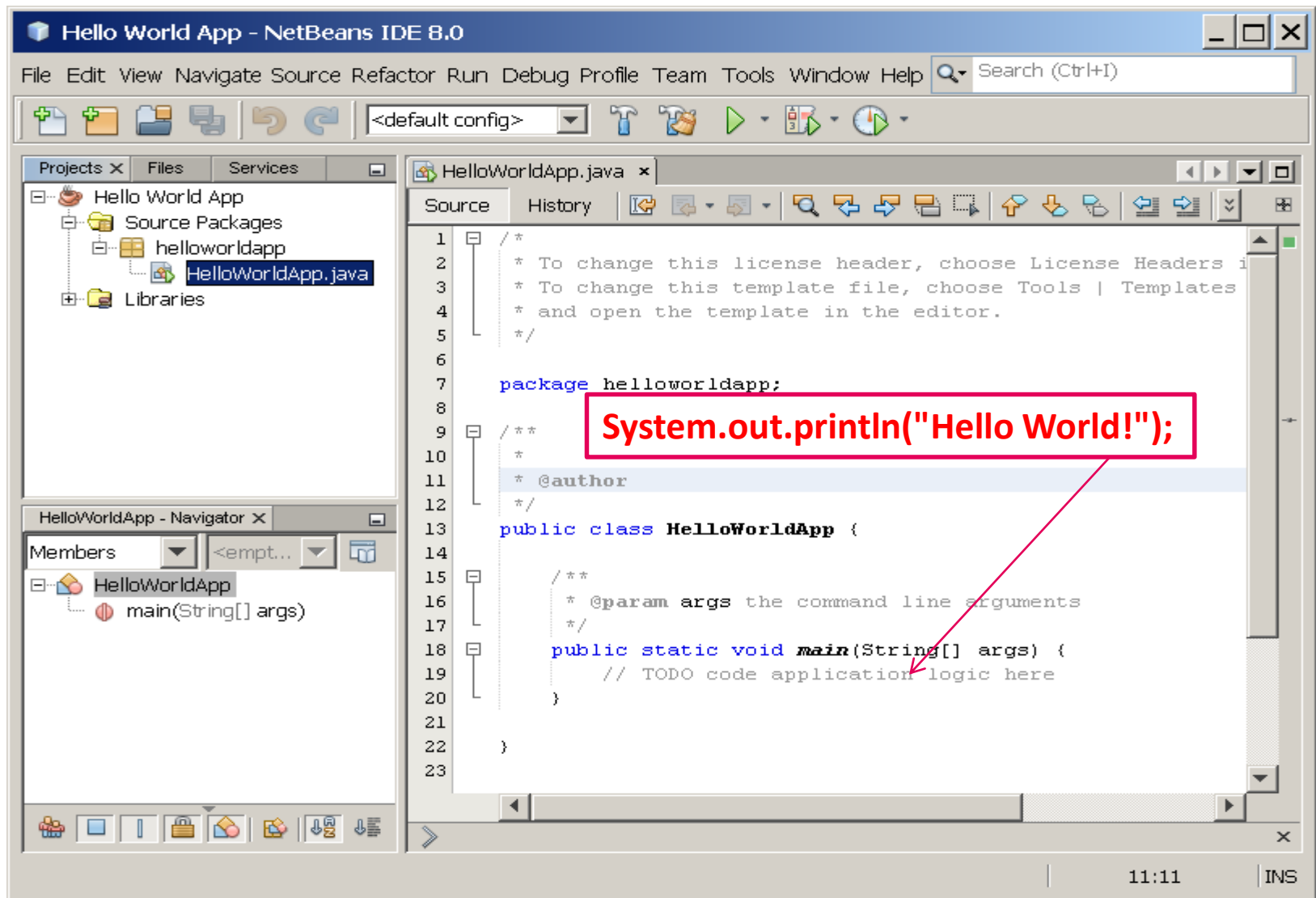
☐ Use Dedicated Folder for Storing Libraries

Libraries Folder:

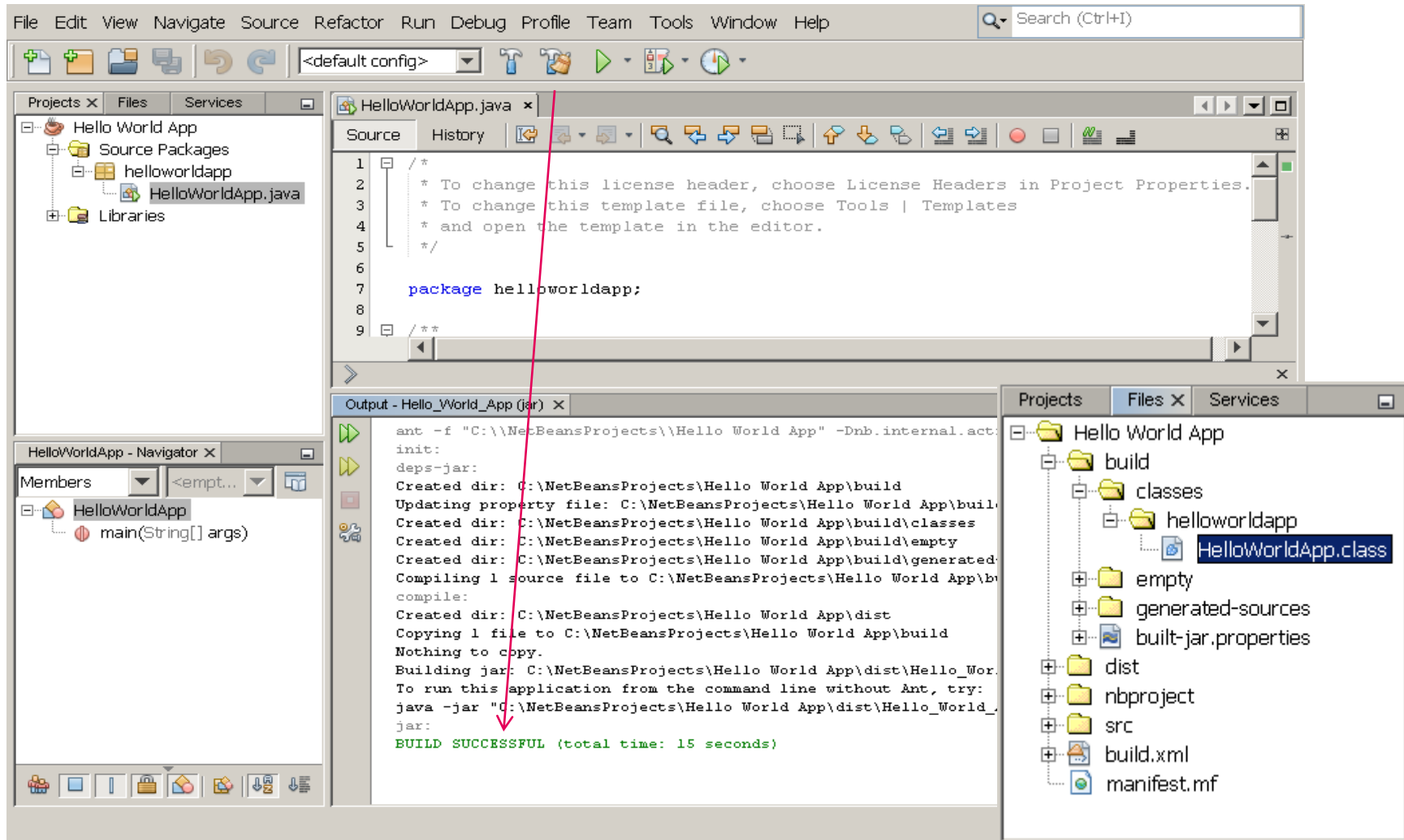
Different users and projects can share the same compilation libraries (see Help for details).

☒ Create Main Class

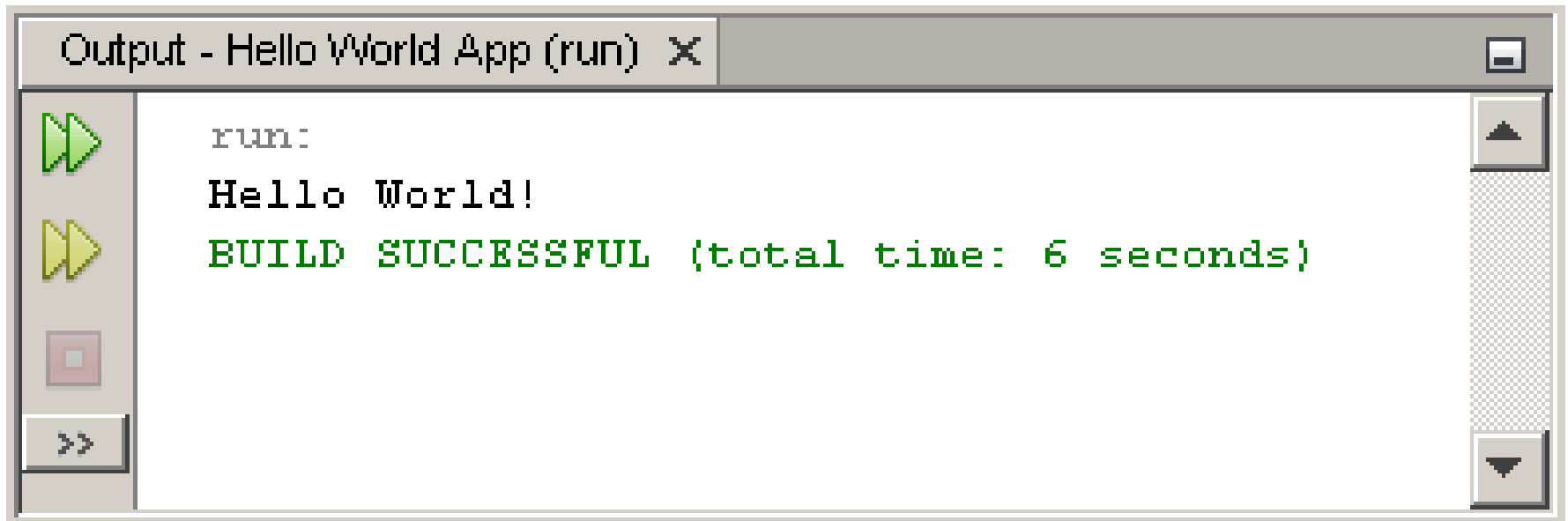
Add Code to the Source File



Compile the Source File into a *.class* File



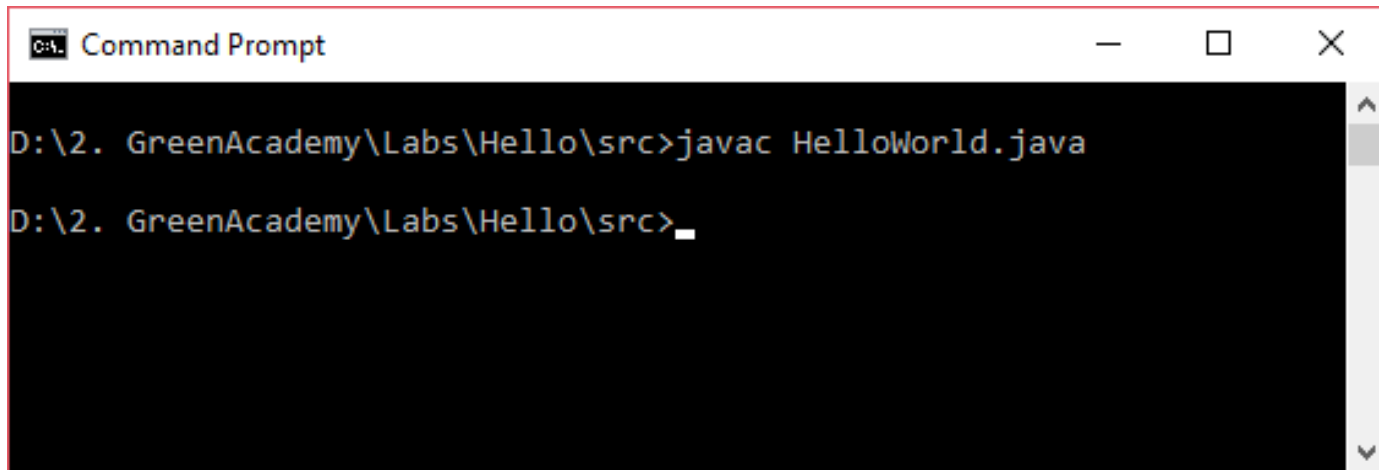
Run the Program



End users run Java Programs

- Set up Environment Variables: PATH and CLASSPATH
- Compile the source file into a .class file
- Create .BAT file
- Run .BAT file

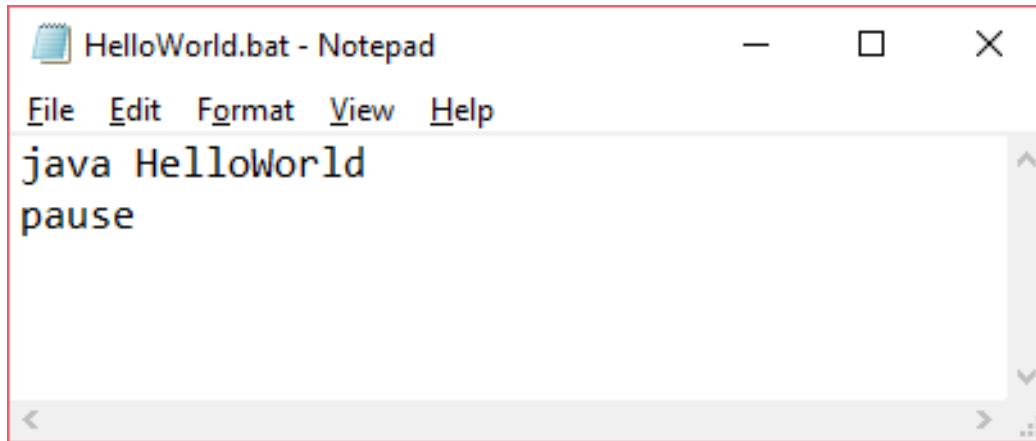
Compile the source file into a .class file



```
Command Prompt
D:\2. GreenAcademy\Labs\Hello\src>javac HelloWorld.java
D:\2. GreenAcademy\Labs\Hello\src>_
```

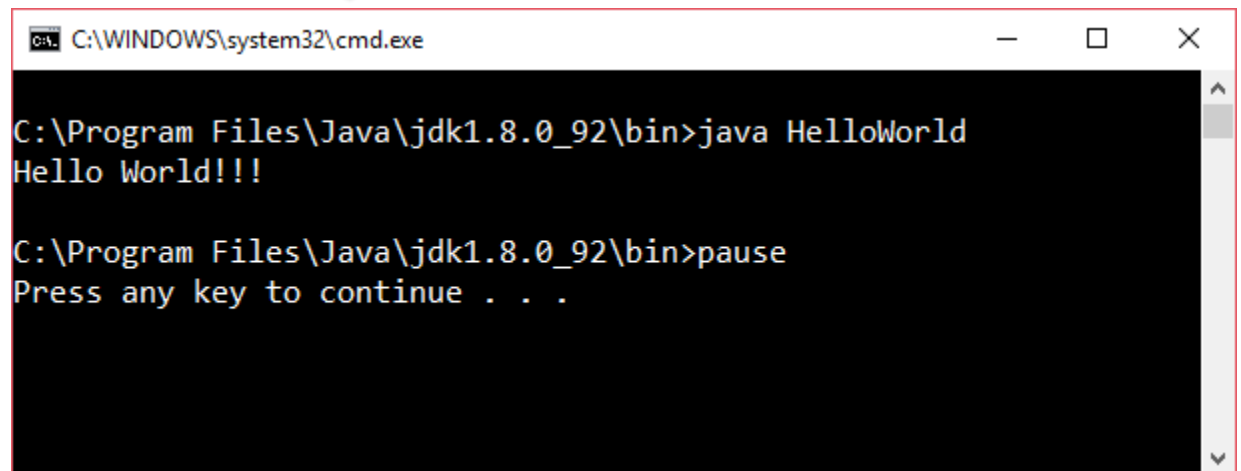
The image shows a Windows Command Prompt window with a black background and white text. The title bar at the top reads "Command Prompt" and includes standard window controls (minimize, maximize, close). The command prompt shows the current directory as "D:\2. GreenAcademy\Labs\Hello\src". The user has entered the command "javac HelloWorld.java", which has been executed. The prompt is now waiting for the next command, indicated by a white cursor after the underscore character.

HelloWorld.bat



A Notepad window titled "HelloWorld.bat - Notepad" with a menu bar (File, Edit, Format, View, Help). The text area contains two lines of code: `java HelloWorld` and `pause`. A large grey arrow points from this window down to the command prompt window below.

```
java HelloWorld
pause
```



A Windows Command Prompt window titled "C:\WINDOWS\system32\cmd.exe" with a black background and white text. It shows the execution of the script from the directory `C:\Program Files\Java\jdk1.8.0_92\bin>`. The first command `java HelloWorld` outputs `Hello World!!!`. The second command `pause` outputs `Press any key to continue . . .`.

```
C:\Program Files\Java\jdk1.8.0_92\bin>java HelloWorld
Hello World!!!

C:\Program Files\Java\jdk1.8.0_92\bin>pause
Press any key to continue . . .
```

Input/Output Data

```
1
2  import java.util.Scanner;
3
4  public class HelloWorld {
5      public static void main(String[] args) {
6          System.out.println("Hello World!!!");
7
8          Scanner sc = new Scanner(System.in);
9          System.out.println("Enter your name: ");
10         String name = sc.nextLine();
11
12         System.out.println("Enter your year of birth: ");
13         int yob = Integer.parseInt(sc.nextLine());
14
15         System.out.println("Hello " + name);
16     }
17 }
18
```